R Notebook

```
library(BSDA)
```

```
## Loading required package: lattice
```

```
##
## Attaching package: 'BSDA'
```

```
## The following object is masked from 'package:datasets':
##
##     Orange
```

```
library(coin)
```

```
## Loading required package: survival
```

```
# Create data
words <- c(15, 7, 12, 14, 11, 10, 4 ,13, 9 , 2)
# order the data in numerical order
word_data <- sort(words)
# mean of data
mean_words <- mean(word_data)
# round 4 digits
mean_words4 <- round(mean_words, 4)
# standard deviation
sd_words <- sd(word_data)
# round 4 digits
sd_words4 <- round(sd_words, 4)
# median of the words
median_words <- median(word_data)
# round 4 digits
median_words4 <- round(median_words, 4)

# dataframe
summarydf <- data.frame(Summary = c("Mean", "Standard Deviation", "Median"),
                        Data = c(mean_words4, sd_words4, median_words4))
summarydf
```

```
##              Summary    Data
## 1               Mean  9.7000
## 2 Standard Deviation  4.2701
## 3             Median 10.5000
```

```r
# summary of word data
summary(word_data)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    2.00    7.50   10.50    9.70   12.75   15.00
```

```r
# Confidence interval for the median (two sided for Confidence intervals)
ci_words <-  SIGN.test(word_data, md = 4, alternative = "two.sided",conf.level = 0.95)
ci_words
```

```
##
##  One-sample Sign-Test
##
## data:  word_data
## s = 8, p-value = 0.03906
## alternative hypothesis: true median is not equal to 4
## 95 percent confidence interval:
##   4.973333 13.675556
## sample estimates:
## median of x
##        10.5
##
## Achieved and Interpolated Confidence Intervals:
##
##                   Conf.Level L.E.pt  U.E.pt
## Lower Achieved CI     0.8906 7.0000 13.0000
## Interpolated CI       0.9500 4.9733 13.6756
## Upper Achieved CI     0.9785 4.0000 14.0000
```

```r
# It's above 4, so 18 year olds better
ci_words$Confidence.Intervals
```

```
##                   Conf.Level L.E.pt  U.E.pt
## Lower Achieved CI     0.8906 7.0000 13.0000
## Interpolated CI       0.9500 4.9733 13.6756
## Upper Achieved CI     0.9785 4.0000 14.0000
```

```r
p_words <- SIGN.test(word_data, md = 4, alternative = "greater",conf.level = 0.95)
p_words
```

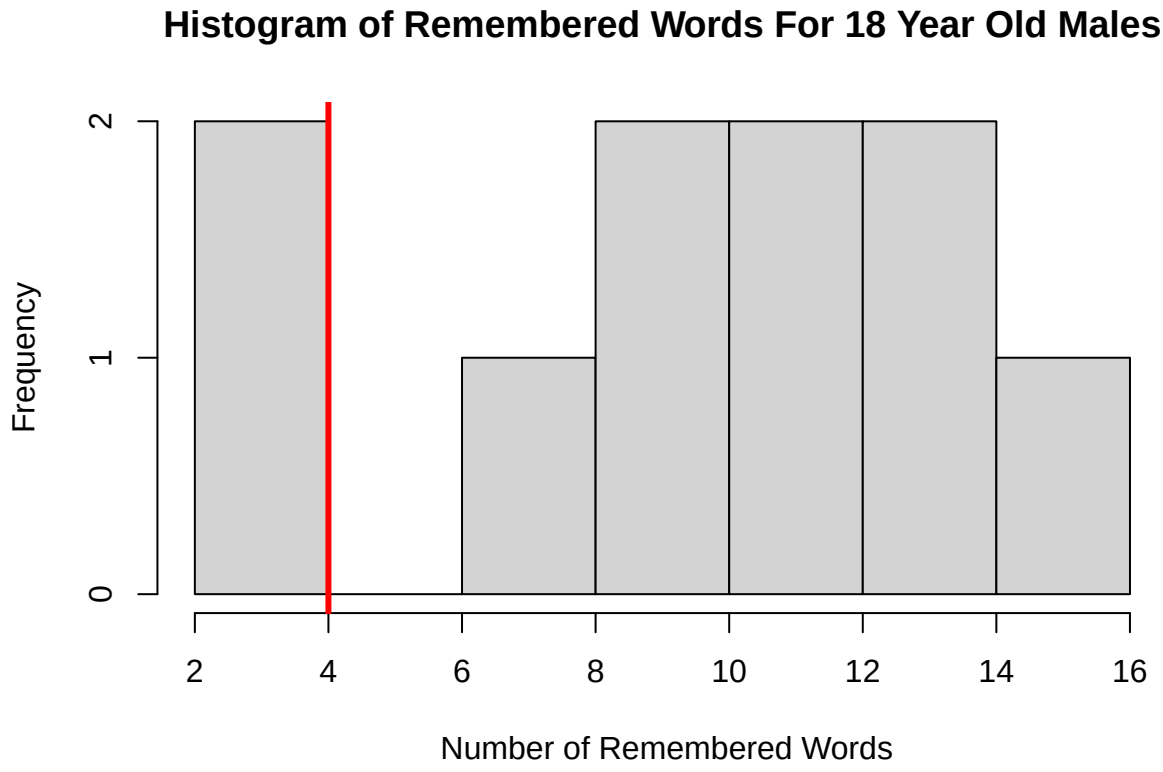
```
##
##  One-sample Sign-Test
##
## data:  word_data
## s = 8, p-value = 0.01953
## alternative hypothesis: true median is greater than 4
## 95 percent confidence interval:
##  6.68  Inf
## sample estimates:
## median of x
##        10.5
```

```
##
## Achieved and Interpolated Confidence Intervals:
##
##                   Conf.Level L.E.pt U.E.pt
## Lower Achieved CI     0.9453   7.00    Inf
## Interpolated CI       0.9500   6.68    Inf
## Upper Achieved CI     0.9893   4.00    Inf
```

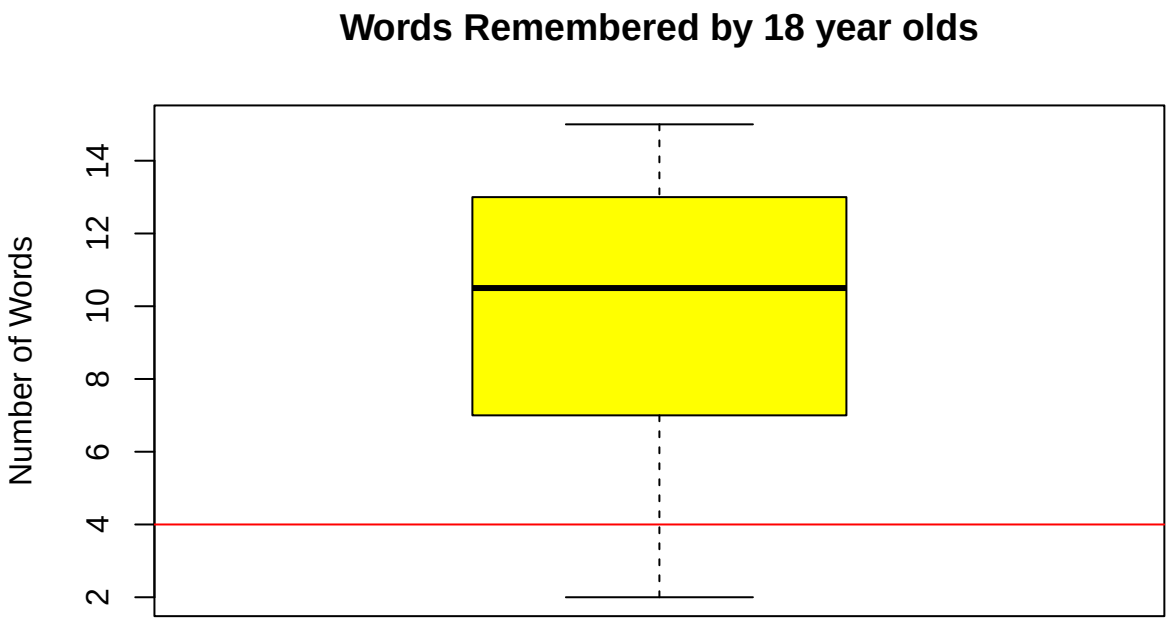
```
# p-value is lower reject null 18 year olds better
p_words$p.value
```

```
## [1] 0.01953125
```

```
#Histogram
hist(word_data, xlab = " Number of Remembered Words", main = "Histogram of Remembered Words For 18 Year
# Red line for 65 year old males
abline(v = 4, col = 'red', lwd = 3)
```




```
#Boxplot
boxplot(word_data, main = "Words Remembered by 18 year olds", ylab = "Number of Words", col = "yellow")
# Line for the median of 65 year old males
abline(h = 4, col = "red")
```


Words Remembered by 18 year olds
Number of Words
2
4
6
8
10
12
14